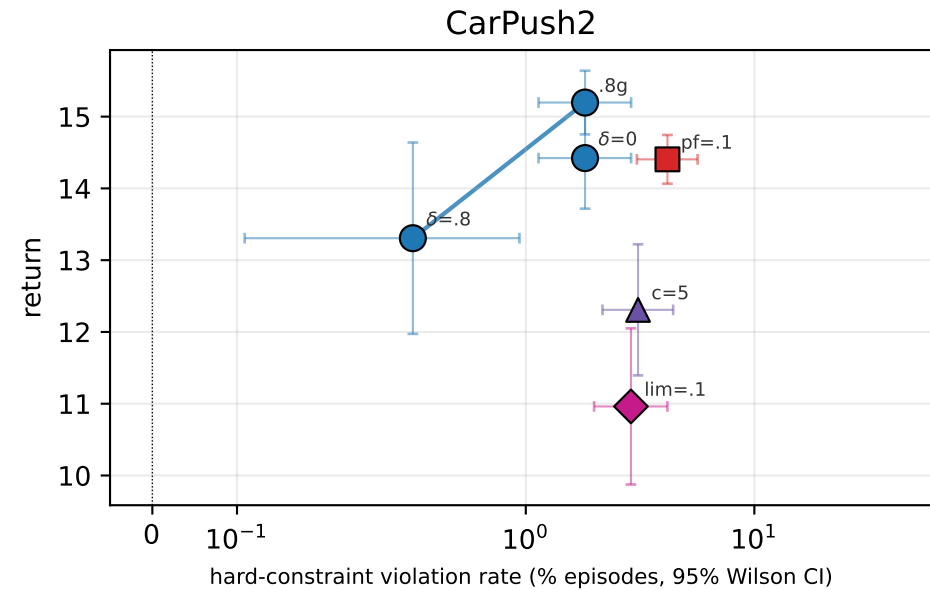
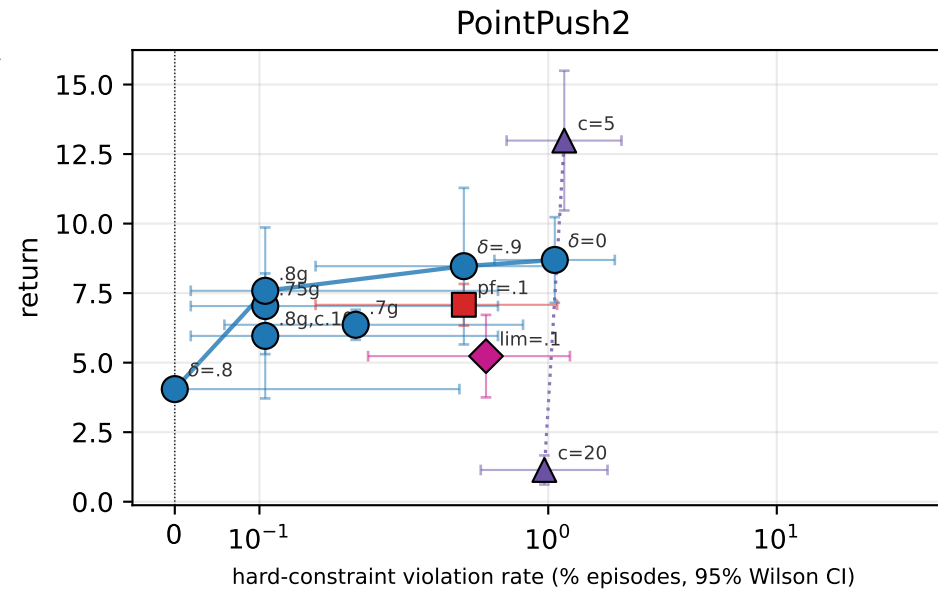


Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)



- SafeOT-typed (ours)
- FDPI (constraint-based)
- ◆ SAC-Lagrangian (penalty-based)
- ▲ SAC + fixed penalty (penalty-based)

- SDAC (multi-constraint SOTA)
- SAC-Pen coefficient sweep
- SRCPO (multi-constraint SOTA)
- FDPI p_f sweep
- typed Pareto frontier (soft ≤ 25)
- SAC-Lag cost-limit sweep

constraint violation rate (% episodes, 95% Wilson CI)